Mei Yi Yang

+1 (734) 417-9376 | mei.yang0508@gmail.com | @meiyy | @mayozoz

Education

University of Michigan — Ann Arbor, MI

Bachelor of Science and Engineering in Computer Science — *August 2023 – May 2027*

- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Web Systems, Machine Learning, Database Management Systems, Computer Organization, Statistics and Data Analytics, Linear Algebra, Game Engines

Experience

AI Engineering Intern — Beijing, China

NetEase YouDao – R&D Team — *May 2025 – August 2025*

Technologies: PyTorch, Transformers/Hugging Face, Python, GPT-4/LLM APIs, CUDA, Pandas/NumPy

- Developed machine translation quality estimation framework for Tibetan-Chinese language pairs, implementing state-of-the-art metrics (COMET, chrF++, BERTScore, NLLB-200) to benchmark translation quality across 1,000+ test samples
- Built automated LLM evaluation system using GPT-4 and Deepseek-v3 models, achieving 78% average correlation with expert annotations and reducing manual review time by 90% through prompt engineering
- Designed statistical validation framework using correlation analysis (Pearson, Spearman, Kendall's tau) to benchmark automated metrics against human judgments, identifying translation model weaknesses with 75% accuracy
- Engineered data processing pipeline for multilingual datasets, implementing custom Chinese tokenization algorithms that improved score reliability by 23% compared to baseline metrics

Frontend Research Assistant — Ann Arbor, MI

Jordan Shavit's Lab — *Febuary 2025 – June 2025*

Technologies: React, JavaScript, HTML, CSS, GitHub Pages

- Developed React-based web application for zebrafish thrombosis image analysis with drag-and-drop interface, deployed on GitHub Pages and used by researchers across 3 collaborating institutions
- Implemented customizable image processing pipeline with adjustable detection parameters and automated batch processing, enabling simultaneous analysis of 50+ images with CSV export for statistical analysis

Operations Team Intern — Shenzhen, China

Tencent Cloud – CSIG Cloud Product Department — *May 2024 – July 2024*

Technologies: Python, Tencent HunYuan AI API, SQL

- Analyzed profitability data for international cloud products (CVM, Lighthouse, GPU) across 8 regions, developing pricing optimization strategies that improved gross margins by 12-18% for underperforming SKUs
- Conducted competitive analysis of AWS, Azure, and GCP pricing across 50+ instance types, identifying opportunities to offer equivalent performance at 5-10% lower cost while maintaining positive margins
- Deployed Tencent HunYuan AI chatbot for internal operations team, automating pricing calculations and reducing manual workload by 80%, saving 15 hours per week across 6 team members

Projects

Thread Library | *C++, Multi-threading, Context Management, Unix*

- Implemented a kernel C++ thread library on Unix, handling CPU booting, thread management, management of 50+ CPUs, interrupts, atomicity, and FIFO scheduling order. Designed spin-locks, mutexes, conditional variables utilizing advanced Unix context management

Virtual Memory Pager | *C++, Virtual Memory, Process Lifecycle Management*

- Designed a virtual memory pager which managed multiple processes and supported swap-backed and file-backed memory pages (similar to mmap()). Managed process creation/forking/destruction, page faults, MMU bits, and swap disk all while supporting copy-on-write.

Fashion Retail Price Tracker | *JavaScript, Chrome Extensions, Canvas API*

- Built a Chrome extension that tracks real-time price history on luxury fashion platforms, featuring a Canvas-rendered sparkline chart with MSRP reference lines, episode-aware restock detection, and a recommendation engine combining linear regression with rule-based heuristics
- Engineered a JSON-LD structured data pipeline for stable price extraction on React SPAs, replacing brittle CSS selectors, with a storage abstraction layer and native ES module architecture requiring zero build tooling

Skills

Languages: C++, Python, JavaScript, Java, C, HTML/CSS, SQL
Frameworks & Libraries: React, Flask, PyTorch, Transformers, NumPy, Pandas, Jinja2
Tools & Platforms: Git, Linux/Bash, AWS S3, GitHub Pages, GDB, CUDA
Design Tools: Blender, Adobe Photoshop, GIMP, Aseprite